Feature Engineering

Procedure:

- Import the required Python libraries such as Pandas and NumPy.
- Load the Titanic dataset and examine its structure and missing values.
- Handle missing values in the dataset.
- Apply one-hot encoding to categorical variables.
- Apply ordinal encoding to variables such as passenger class.
- Perform count encoding and frequency encoding on categorical features.
- Apply target encoding, mean encoding, and probability-ratio encoding.
- Display the transformed dataset and verify the results.

Solution:

```
# Feature Engineering: Handle Categorical Data

import pandas as pd
import numpy as np

# Load the dataset
dt = pd.read_csv(r"C:\Users\Lenovo\Downloads\BHAGYASHREE BAGALUR SDM
College(Autonomous), Ujire - train_FE (1).csv")

# Display dataset information
dt.info()

# Check for null values
print(dt.isnull().sum())


# ------------------------------------------------
# 1. HANDLE CONTINUOUS MISSING DATA
# ------------------------------------------------

# Function to impute missing values
def impute_nan(DataFrame, ColumnName, ImputeValue):
    DataFrame[ColumnName] = DataFrame[ColumnName].fillna(ImputeValue)

# Impute median value for Age
median = dt['Age'].median()
impute_nan(dt, 'Age', median)

# Check null values
print(dt.isnull().sum())
```

```
# ------------------------------------------------
# 2. HANDLE CATEGORICAL MISSING DATA USING MODE
# ------------------------------------------------

# Function to replace NaN values with the most frequent category
def impute_nan_most_frequent_category(DF, ColName):
    most_frequent_category = DF[ColName].mode()[0]
    DF[ColName] = DF[ColName].fillna(most_frequent_category)

# Impute missing categorical values
for Columns in ['Embarked', 'Cabin_Serial', 'Cabin']:
    impute_nan_most_frequent_category(dt, Columns)

# Check null values
print(dt.isnull().sum())


# ------------------------------------------------
# 3. CREATE DUMMY VARIABLES (ONE-HOT ENCODING)
# ------------------------------------------------

dt_dummies = pd.get_dummies(dt)

print(dt_dummies.head())


# ------------------------------------------------
# 4. ORDINAL NUMBER ENCODING
# ------------------------------------------------

# Create a dictionary with category names and their ranks
PClassDict = {
    'First': 1,
    'Second': 2,
    'Third': 3
}

# Create a new column using the dictionary
dt['Ord_Pclass'] = dt['Pclass'].map(PClassDict)

print(dt['Ord_Pclass'].unique())

# Display the result
print(dt[['Pclass', 'Ord_Pclass']].head(10))

# Drop the original column
dt = dt.drop('Pclass', axis=1)

print(dt.head(10))


# ------------------------------------------------
```

```
# 5. COUNT / FREQUENCY ENCODING
# ------------------------------------------------

dt2 = dt.copy()

print(dt2.shape)

# Create a list of categorical columns
dt_cat = dt.select_dtypes(
    include=['object', 'category']
).columns.tolist()

print(dt_cat)

# Create dictionaries containing category frequencies
Salutation_Dict = dt2['Salutation'].value_counts()
Sex_Dict = dt2['Sex'].value_counts()
Embarked_Dict = dt2['Embarked'].value_counts()
Cabin_Serial_Dict = dt2['Cabin_Serial'].value_counts()
Cabin_Dict = dt2['Cabin'].value_counts()

print(Salutation_Dict)

# Create new columns using frequency mapping
dt2['Salutation_Dict'] = dt2['Salutation'].map(Salutation_Dict)
dt2['Sex_Dict'] = dt2['Sex'].map(Sex_Dict)
dt2['Embarked_Dict'] = dt2['Embarked'].map(Embarked_Dict)
dt2['Cabin_Serial_Dict'] = dt2['Cabin_Serial'].map(Cabin_Serial_Dict)
dt2['Cabin_Dict'] = dt2['Cabin'].map(Cabin_Dict)

# Display the result
print(dt2[
    ['Salutation', 'Salutation_Dict',
     'Sex', 'Sex_Dict',
     'Embarked', 'Embarked_Dict',
     'Cabin_Serial', 'Cabin_Serial_Dict',
     'Cabin', 'Cabin_Dict']
].head())

# Drop the original categorical columns
dt2 = dt2.drop(
    ['Salutation', 'Sex', 'Embarked', 'Cabin_Serial', 'Cabin'],
    axis=1
)

print(dt2.head())


# ------------------------------------------------
# 6. TARGET / GUIDED ENCODING
# ------------------------------------------------
```

```
dt3 = dt.copy()

# Take the first character of Cabin
dt3['Cabin'] = dt3['Cabin'].astype(str).str[0]

# Calculate the survival probability for each cabin category
print(dt3.groupby(['Cabin'])['Survived'].mean())

# Calculate the overall survival probability
print(dt3['Survived'].mean())

# Sort categories by their survival probabilities
Encoded_Lables = dt3.groupby(
    ['Cabin']
)['Survived'].mean().sort_values().index

print(Encoded_Lables)

# Create a dictionary with probability rankings
Encoded_Lables_Ranks = {
    k: i for i, k in enumerate(Encoded_Lables, 0)
}

print(Encoded_Lables_Ranks)

# Map cabin categories to their probability rankings
dt3['Cabin_Encoded'] = dt3['Cabin'].map(Encoded_Lables_Ranks)

# Display the result
print(dt3[['Cabin', 'Cabin_Encoded']].head(10))

# Drop the original Cabin column
dt3 = dt3.drop('Cabin', axis=1)


# ------------------------------------------------
# 7. MEAN ENCODING
# ------------------------------------------------

dt4 = dt.copy()

# Calculate the mean survival rate for each cabin category
Encoded_Mean_Dict = dt4.groupby(
    ['Cabin']
)['Survived'].mean().to_dict()

# Replace cabin categories with their mean survival rates
dt4['Cabin_Mean_Encoded'] = dt4['Cabin'].map(Encoded_Mean_Dict)

# Display the result
```

```
print(dt4[['Cabin', 'Cabin_Mean_Encoded']].head(10))

# Drop the original Cabin column
dt4 = dt4.drop('Cabin', axis=1)


# ------------------------------------------------
# 8. PROBABILITY RATIO ENCODING
# ------------------------------------------------

dt5 = dt.copy()

# Replace Cabin values with their first character
dt5['Cabin'] = dt5['Cabin'].astype(str).str[0]

# Calculate the probability of survival for each cabin category
Probability_Survived = dt5.groupby(
    ['Cabin']
)['Survived'].mean()

Probability_Survived = pd.DataFrame(Probability_Survived)

# Calculate the probability of death
Probability_Survived['Died'] = (
    1 - Probability_Survived['Survived']
)

print(Probability_Survived)

# Calculate the ratio of survival probability to death probability
Probability_Survived['Prob_Ratio'] = (
    Probability_Survived['Survived'] /
    Probability_Survived['Died']
)

print(Probability_Survived)

# Create a dictionary containing probability ratios
Encode_Prob_Ratio = Probability_Survived['Prob_Ratio'].to_dict()

print(Encode_Prob_Ratio)

# Create a new column using probability ratio encoding
dt5['Encode_Prob_Ratio'] = dt5['Cabin'].map(Encode_Prob_Ratio)

# Display the result
print(dt5[['Cabin', 'Encode_Prob_Ratio']].head(10))

# Drop the original Cabin column
dt5 = dt5.drop('Cabin', axis=1)
```

print(dt5.head(10))

Output:

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 14 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   PassengerId   891 non-null    int64
 1   Survived      891 non-null    int64
 2   Pclass        891 non-null    str
 3   Salutation    891 non-null    str
 4   Name          891 non-null    str
 5   Sex           891 non-null    str
 6   Age           714 non-null    float64
 7   SibSp         891 non-null    int64
 8   Parch         891 non-null    int64
 9   Ticket        891 non-null    str
 10  Fare          891 non-null    float64
 11  Embarked      889 non-null    str
 12  Cabin_Serial  204 non-null    str
 13  Cabin         204 non-null    str
dtypes: float64(2), int64(4), str(8)
memory usage: 97.6 KB
PassengerId       0
Survived          0
Pclass            0
Salutation        0
Name              0
Sex               0
Age             177
SibSp             0
Parch             0
Ticket            0
Fare              0
Embarked          2
Cabin_Serial    687
Cabin           687
dtype: int64
PassengerId       0
Survived          0
Pclass            0
Salutation        0
Name              0
Sex               0
Age               0
SibSp             0
Parch             0
Ticket            0
Fare              0
Embarked          2
Cabin_Serial    687
Cabin           687
```

```
dtype: int64
PassengerId     0
Survived        0
Pclass          0
Salutation      0
Name            0
Sex             0
Age             0
SibSp           0
Parch           0
Ticket          0
Fare            0
Embarked        0
Cabin_Serial    0
Cabin           0
dtype: int64
   PassengerId  Survived   Age  SibSp  Parch     Fare  Pclass_First  ...  Cabin_F G73
Cabin_F2  Cabin_F33  Cabin_F38  Cabin_F4  Cabin_G6  Cabin_T
0            1         0  22.0      1      0   7.2500         False  ...        False
False      False      False     False     False    False
1            2         1  38.0      1      0  71.2833          True  ...        False
False      False      False     False     False    False
2            3         1  26.0      0      0   7.9250         False  ...        False
False      False      False     False     False    False
3            4         1  35.0      1      0  53.1000          True  ...        False
False      False      False     False     False    False
4            5         0  35.0      0      0   8.0500         False  ...        False
False      False      False     False     False    False

[5 rows x 1745 columns]
[3 1 2]
   Pclass  Ord_Pclass
0   Third           3
1   First           1
2   Third           3
3   First           1
4   Third           3
5   Third           3
6   First           1
7   Third           3
8   Third           3
9  Second           2
   PassengerId  Survived Salutation                                                Name
Sex  ...     Fare  Embarked  Cabin_Serial    Cabin  Ord_Pclass
0            1         0         Mr                              Braund, Mr. Owen Harris
male  ...   7.2500         S             C  B96 B98           3
1            2         1        Mrs  Cumings, Mrs. John Bradley (Florence Briggs Th...
female  ...  71.2833         C             C      C85           1
2            3         1       Miss                               Heikkinen, Miss. Laina
female  ...   7.9250         S             C  B96 B98           3
3            4         1        Mrs       Futrelle, Mrs. Jacques Heath (Lily May Peel)
female  ...  53.1000         S             C     C123           1
4            5         0         Mr                             Allen, Mr. William Henry
male  ...   8.0500         S             C  B96 B98           3
5            6         0         Mr                                    Moran, Mr. James
male  ...   8.4583         Q             C  B96 B98           3
```

```
6            7         0         Mr                              McCarthy, Mr. Timothy J
male  ...  51.8625         S             E      E46           1
7            8         0     Master                      Palsson, Master. Gosta Leonard
male  ...  21.0750         S             C  B96 B98           3
8            9         1        Mrs  Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)
female  ...  11.1333         S             C  B96 B98           3
9           10         1        Mrs                Nasser, Mrs. Nicholas (Adele Achem)
female  ...  30.0708         C             C  B96 B98           2

[10 rows x 14 columns]
(891, 14)
c:\Users\Lenovo\Desktop\Python Prgm\prac10.py:92: Pandas4Warning: For backward
compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified.
This behavior is deprecated and will be removed in a future version. Explicitly pass 'str'
to `include` to select them, or to `exclude` to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-
select-dtypes for details on how to write code that works with pandas 2 and 3.
  dt_cat = dt.select_dtypes(
['Salutation', 'Name', 'Sex', 'Ticket', 'Embarked', 'Cabin_Serial', 'Cabin']
Salutation
Mr        537
Miss      185
Mrs       129
Master     40
Name: count, dtype: int64
  Salutation  Salutation_Dict     Sex  Sex_Dict Embarked  Embarked_Dict Cabin_Serial
Cabin_Serial_Dict    Cabin  Cabin_Dict
0         Mr              537    male       577        S            646           C
746  B96 B98         691
1        Mrs              129  female       314        C            168           C
746      C85           1
2       Miss              185  female       314        S            646           C
746  B96 B98         691
3        Mrs              129  female       314        S            646           C
746     C123           2
4         Mr              537    male       577        S            646           C
746  B96 B98         691
   PassengerId  Survived                                          Name   Age  SibSp
...  Salutation_Dict Sex_Dict  Embarked_Dict  Cabin_Serial_Dict  Cabin_Dict
0            1         0                             Braund, Mr. Owen Harris  22.0      1
...              537      577            646                746         691
1            2         1  Cumings, Mrs. John Bradley (Florence Briggs Th...  38.0      1
...              129      314            168                746           1
2            3         1                              Heikkinen, Miss. Laina  26.0      0
...              185      314            646                746         691
3            4         1       Futrelle, Mrs. Jacques Heath (Lily May Peel)  35.0      1
...              129      314            646                746           2
4            5         0                            Allen, Mr. William Henry  35.0      0
...              537      577            646                746         691

[5 rows x 14 columns]
Cabin
A    0.466667
B    0.328338
C    0.593220
D    0.757576
```

```
E    0.750000
F    0.615385
G    0.500000
T    0.000000
Name: Survived, dtype: float64
0.3838383838383838
Index(['T', 'B', 'A', 'G', 'C', 'F', 'E', 'D'], dtype='str', name='Cabin')
{'T': 0, 'B': 1, 'A': 2, 'G': 3, 'C': 4, 'F': 5, 'E': 6, 'D': 7}
  Cabin  Cabin_Encoded
0     B              1
1     C              4
2     B              1
3     C              4
4     B              1
5     B              1
6     E              6
7     B              1
8     B              1
9     B              1
     Cabin  Cabin_Mean_Encoded
0  B96 B98            0.303907
1      C85            1.000000
2  B96 B98            0.303907
3     C123            0.500000
4  B96 B98            0.303907
5  B96 B98            0.303907
6      E46            0.000000
7  B96 B98            0.303907
8  B96 B98            0.303907
9  B96 B98            0.303907
       Survived      Died
Cabin
A      0.466667  0.533333
B      0.328338  0.671662
C      0.593220  0.406780
D      0.757576  0.242424
E      0.750000  0.250000
F      0.615385  0.384615
G      0.500000  0.500000
T      0.000000  1.000000
       Survived      Died  Prob_Ratio
Cabin
A      0.466667  0.533333    0.875000
B      0.328338  0.671662    0.488844
C      0.593220  0.406780    1.458333
D      0.757576  0.242424    3.125000
E      0.750000  0.250000    3.000000
F      0.615385  0.384615    1.600000
G      0.500000  0.500000    1.000000
T      0.000000  1.000000    0.000000
{'A': 0.875, 'B': 0.488843813387424, 'C': 1.4583333333333333, 'D': 3.125, 'E': 3.0, 'F':
1.6000000000000003, 'G': 1.0, 'T': 0.0}
  Cabin  Encode_Prob_Ratio
0     B           0.488844
1     C           1.458333
```

```
2     B           0.488844
3     C           1.458333
4     B           0.488844
5     B           0.488844
6     E           3.000000
7     B           0.488844
8     B           0.488844
9     B           0.488844
   PassengerId  Survived Salutation                                                 Name
Sex  ...     Fare  Embarked  Cabin_Serial Ord_Pclass  Encode_Prob_Ratio
0            1         0         Mr                            Braund, Mr. Owen Harris
male  ...   7.2500         S             C          3           0.488844
1            2         1        Mrs  Cumings, Mrs. John Bradley (Florence Briggs Th...
female  ...  71.2833         C             C          1           1.458333
2            3         1       Miss                            Heikkinen, Miss. Laina
female  ...   7.9250         S             C          3           0.488844
3            4         1        Mrs       Futrelle, Mrs. Jacques Heath (Lily May Peel)
female  ...  53.1000         S             C          1           1.458333
4            5         0         Mr                           Allen, Mr. William Henry
male  ...   8.0500         S             C          3           0.488844
5            6         0         Mr                                 Moran, Mr. James
male  ...   8.4583         Q             C          3           0.488844
6            7         0         Mr                            McCarthy, Mr. Timothy J
male  ...  51.8625         S             E          1           3.000000
7            8         0     Master                        Palsson, Master. Gosta Leonard
male  ...  21.0750         S             C          3           0.488844
8            9         1        Mrs  Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)
female  ...  11.1333         S             C          3           0.488844
9           10         1        Mrs                Nasser, Mrs. Nicholas (Adele Achem)
female  ...  30.0708         C             C          2           0.488844

[10 rows x 14 columns]
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 14 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   PassengerId   891 non-null    int64
 1   Survived      891 non-null    int64
 2   Pclass        891 non-null    str
 3   Salutation    891 non-null    str
 4   Name          891 non-null    str
 5   Sex           891 non-null    str
 6   Age           714 non-null    float64
 7   SibSp         891 non-null    int64
 8   Parch         891 non-null    int64
 9   Ticket        891 non-null    str
 10  Fare          891 non-null    float64
 11  Embarked      889 non-null    str
 12  Cabin_Serial  204 non-null    str
 13  Cabin         204 non-null    str
dtypes: float64(2), int64(4), str(8)
memory usage: 97.6 KB
PassengerId       0
Survived          0
Pclass            0
```

```
Salutation        0
Name              0
Sex               0
Age             177
SibSp             0
Parch             0
Ticket            0
Fare              0
Embarked          2
Cabin_Serial    687
Cabin           687
dtype: int64
PassengerId       0
Survived          0
Pclass            0
Salutation        0
Name              0
Sex               0
Age               0
SibSp             0
Parch             0
Ticket            0
Fare              0
Embarked          2
Cabin_Serial    687
Cabin           687
dtype: int64
PassengerId     0
Survived        0
Pclass          0
Salutation      0
Name            0
Sex             0
Age             0
SibSp           0
Parch           0
Ticket          0
Fare            0
Embarked        0
Cabin_Serial    0
Cabin           0
dtype: int64
   PassengerId  Survived   Age  SibSp  Parch     Fare  Pclass_First  ...  Cabin_F G73
Cabin_F2  Cabin_F33  Cabin_F38  Cabin_F4  Cabin_G6  Cabin_T
0            1         0  22.0      1      0   7.2500         False  ...        False
False      False      False     False     False    False
1            2         1  38.0      1      0  71.2833          True  ...        False
False      False      False     False     False    False
2            3         1  26.0      0      0   7.9250         False  ...        False
False      False      False     False     False    False
3            4         1  35.0      1      0  53.1000          True  ...        False
False      False      False     False     False    False
4            5         0  35.0      0      0   8.0500         False  ...        False
False      False      False     False     False    False
```

```
[5 rows x 1745 columns]
[3 1 2]
   Pclass  Ord_Pclass
0   Third           3
1   First           1
2   Third           3
3   First           1
4   Third           3
5   Third           3
6   First           1
7   Third           3
8   Third           3
9  Second           2
   PassengerId  Survived Salutation                                                 Name
Sex  ...     Fare  Embarked  Cabin_Serial    Cabin  Ord_Pclass
0            1         0         Mr                             Braund, Mr. Owen Harris
male  ...   7.2500         S             C  B96 B98           3
1            2         1        Mrs  Cumings, Mrs. John Bradley (Florence Briggs Th...
female  ...  71.2833         C             C      C85           1
2            3         1       Miss                              Heikkinen, Miss. Laina
female  ...   7.9250         S             C  B96 B98           3
3            4         1        Mrs       Futrelle, Mrs. Jacques Heath (Lily May Peel)
female  ...  53.1000         S             C     C123           1
4            5         0         Mr                            Allen, Mr. William Henry
male  ...   8.0500         S             C  B96 B98           3
5            6         0         Mr                                    Moran, Mr. James
male  ...   8.4583         Q             C  B96 B98           3
6            7         0         Mr                             McCarthy, Mr. Timothy J
male  ...  51.8625         S             E      E46           1
7            8         0     Master                      Palsson, Master. Gosta Leonard
male  ...  21.0750         S             C  B96 B98           3
8            9         1        Mrs  Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)
female  ...  11.1333         S             C  B96 B98           3
9           10         1        Mrs                Nasser, Mrs. Nicholas (Adele Achem)
female  ...  30.0708         C             C  B96 B98           2

[10 rows x 14 columns]
(891, 14)
c:\Users\Lenovo\Desktop\Python Prgm\prac10.py:92: Pandas4Warning: For backward
compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified.
This behavior is deprecated and will be removed in a future version. Explicitly pass 'str'
to `include` to select them, or to `exclude` to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-
select-dtypes for details on how to write code that works with pandas 2 and 3.
  dt_cat = dt.select_dtypes(
['Salutation', 'Name', 'Sex', 'Ticket', 'Embarked', 'Cabin_Serial', 'Cabin']
Salutation
Mr        537
Miss      185
Mrs       129
Master     40
Name: count, dtype: int64
  Salutation  Salutation_Dict     Sex  Sex_Dict Embarked  Embarked_Dict Cabin_Serial
Cabin_Serial_Dict    Cabin  Cabin_Dict
0         Mr              537    male       577        S            646            C
746  B96 B98         691
```

```
1        Mrs              129  female       314        C            168            C
746      C85            1
2       Miss              185  female       314        S            646            C
746  B96 B98          691
3        Mrs              129  female       314        S            646            C
746     C123            2
4         Mr              537    male       577        S            646            C
746  B96 B98          691
   PassengerId  Survived                                              Name   Age  SibSp
...  Salutation_Dict Sex_Dict  Embarked_Dict  Cabin_Serial_Dict  Cabin_Dict
0            1         0                              Braund, Mr. Owen Harris  22.0      1
...              537      577            646                746         691
1            2         1  Cumings, Mrs. John Bradley (Florence Briggs Th...  38.0      1
...              129      314            168                746           1
2            3         1                               Heikkinen, Miss. Laina  26.0      0
...              185      314            646                746         691
3            4         1       Futrelle, Mrs. Jacques Heath (Lily May Peel)  35.0      1
...              129      314            646                746           2
4            5         0                             Allen, Mr. William Henry  35.0      0
...              537      577            646                746         691

[5 rows x 14 columns]
Cabin
A    0.466667
B    0.328338
C    0.593220
D    0.757576
E    0.750000
F    0.615385
G    0.500000
T    0.000000
Name: Survived, dtype: float64
0.3838383838383838
Index(['T', 'B', 'A', 'G', 'C', 'F', 'E', 'D'], dtype='str', name='Cabin')
{'T': 0, 'B': 1, 'A': 2, 'G': 3, 'C': 4, 'F': 5, 'E': 6, 'D': 7}
  Cabin  Cabin_Encoded
0     B              1
1     C              4
2     B              1
3     C              4
4     B              1
5     B              1
6     E              6
7     B              1
8     B              1
9     B              1
     Cabin  Cabin_Mean_Encoded
0  B96 B98           0.303907
1      C85           1.000000
2  B96 B98           0.303907
3     C123           0.500000
4  B96 B98           0.303907
5  B96 B98           0.303907
6      E46           0.000000
7  B96 B98           0.303907
8  B96 B98           0.303907
```

```
9  B96 B98            0.303907
       Survived      Died
Cabin
A      0.466667  0.533333
B      0.328338  0.671662
C      0.593220  0.406780
D      0.757576  0.242424
E      0.750000  0.250000
F      0.615385  0.384615
G      0.500000  0.500000
T      0.000000  1.000000
       Survived      Died  Prob_Ratio
Cabin
A      0.466667  0.533333    0.875000
B      0.328338  0.671662    0.488844
C      0.593220  0.406780    1.458333
D      0.757576  0.242424    3.125000
E      0.750000  0.250000    3.000000
F      0.615385  0.384615    1.600000
G      0.500000  0.500000    1.000000
T      0.000000  1.000000    0.000000
{'A': 0.875, 'B': 0.488843813387424, 'C': 1.4583333333333333, 'D': 3.125, 'E': 3.0, 'F':
1.6000000000000003, 'G': 1.0, 'T': 0.0}
  Cabin  Encode_Prob_Ratio
0     B           0.488844
1     C           1.458333
2     B           0.488844
3     C           1.458333
4     B           0.488844
5     B           0.488844
6     E           3.000000
7     B           0.488844
8     B           0.488844
9     B           0.488844
   PassengerId  Survived Salutation                                          Name
Sex  ...     Fare  Embarked  Cabin_Serial Ord_Pclass  Encode_Prob_Ratio
0            1         0         Mr                              Braund, Mr. Owen Harris
male  ...   7.2500         S             C          3           0.488844
1            2         1        Mrs  Cumings, Mrs. John Bradley (Florence Briggs Th...
female  ...  71.2833         C             C          1           1.458333
2            3         1       Miss                              Heikkinen, Miss. Laina
female  ...   7.9250         S             C          3           0.488844
3            4         1        Mrs       Futrelle, Mrs. Jacques Heath (Lily May Peel)
female  ...  53.1000         S             C          1           1.458333
4            5         0         Mr                             Allen, Mr. William Henry
male  ...   8.0500         S             C          3           0.488844
5            6         0         Mr                                   Moran, Mr. James
male  ...   8.4583         Q             C          3           0.488844
6            7         0         Mr                             McCarthy, Mr. Timothy J
male  ...  51.8625         S             E          1           3.000000
7            8         0     Master                         Palsson, Master. Gosta Leonard
male  ...  21.0750         S             C          3           0.488844
8            9         1        Mrs  Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)
female  ...  11.1333         S             C          3           0.488844
9           10         1        Mrs                Nasser, Mrs. Nicholas (Adele Achem)
female  ...  30.0708         C             C          2           0.488844
```

```
[10 rows x 14 columns]
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 14 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   PassengerId   891 non-null    int64
 1   Survived      891 non-null    int64
 2   Pclass        891 non-null    str
 3   Salutation    891 non-null    str
 4   Name          891 non-null    str
 5   Sex           891 non-null    str
 6   Age           714 non-null    float64
 7   SibSp         891 non-null    int64
 8   Parch         891 non-null    int64
 9   Ticket        891 non-null    str
 10  Fare          891 non-null    float64
 11  Embarked      889 non-null    str
 12  Cabin_Serial  204 non-null    str
 13  Cabin         204 non-null    str
dtypes: float64(2), int64(4), str(8)
memory usage: 97.6 KB
PassengerId       0
Survived          0
Pclass            0
Salutation        0
Name              0
Sex               0
Age             177
SibSp             0
Parch             0
Ticket            0
Fare              0
Embarked          2
Cabin_Serial    687
Cabin           687
dtype: int64
PassengerId       0
Survived          0
Pclass            0
Salutation        0
Name              0
Sex               0
Age               0
SibSp             0
Parch             0
Ticket            0
Fare              0
Embarked          2
Cabin_Serial    687
Cabin           687
dtype: int64
PassengerId     0
Survived        0
```

```
Pclass          0
Salutation      0
Name            0
Sex             0
Age             0
SibSp           0
Parch           0
Ticket          0
Fare            0
Embarked        0
Cabin_Serial    0
Cabin           0
dtype: int64
   PassengerId  Survived   Age  SibSp  Parch     Fare  Pclass_First  ...  Cabin_F G73
Cabin_F2  Cabin_F33  Cabin_F38  Cabin_F4  Cabin_G6  Cabin_T
0            1         0  22.0      1      0   7.2500         False  ...        False
False      False      False     False     False    False
1            2         1  38.0      1      0  71.2833          True  ...        False
False      False      False     False     False    False
2            3         1  26.0      0      0   7.9250         False  ...        False
False      False      False     False     False    False
3            4         1  35.0      1      0  53.1000          True  ...        False
False      False      False     False     False    False
4            5         0  35.0      0      0   8.0500         False  ...        False
False      False      False     False     False    False

[5 rows x 1745 columns]
[3 1 2]
   Pclass  Ord_Pclass
0   Third           3
1   First           1
2   Third           3
3   First           1
4   Third           3
5   Third           3
6   First           1
7   Third           3
8   Third           3
9  Second           2
   PassengerId  Survived Salutation                                                Name
Sex  ...     Fare  Embarked  Cabin_Serial    Cabin  Ord_Pclass
0            1         0         Mr                             Braund, Mr. Owen Harris
male  ...   7.2500         S             C  B96 B98           3
1            2         1        Mrs  Cumings, Mrs. John Bradley (Florence Briggs Th...
female  ...  71.2833         C             C      C85           1
2            3         1       Miss                              Heikkinen, Miss. Laina
female  ...   7.9250         S             C  B96 B98           3
3            4         1        Mrs       Futrelle, Mrs. Jacques Heath (Lily May Peel)
female  ...  53.1000         S             C     C123           1
4            5         0         Mr                            Allen, Mr. William Henry
male  ...   8.0500         S             C  B96 B98           3
5            6         0         Mr                                    Moran, Mr. James
male  ...   8.4583         Q             C  B96 B98           3
6            7         0         Mr                             McCarthy, Mr. Timothy J
male  ...  51.8625         S             E      E46           1
```

```
7            8         0     Master                      Palsson, Master. Gosta Leonard
male  ...  21.0750         S             C  B96 B98           3
8            9         1        Mrs  Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)
female  ...  11.1333         S             C  B96 B98           3
9           10         1        Mrs                Nasser, Mrs. Nicholas (Adele Achem)
female  ...  30.0708         C             C  B96 B98           2

[10 rows x 14 columns]
(891, 14)
c:\Users\Lenovo\Desktop\Python Prgm\prac10.py:92: Pandas4Warning: For backward
compatibility, 'str' dtypes are included by select_dtypes when 'object' dtype is specified.
This behavior is deprecated and will be removed in a future version. Explicitly pass 'str'
to `include` to select them, or to `exclude` to remove them and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-
select-dtypes for details on how to write code that works with pandas 2 and 3.
  dt_cat = dt.select_dtypes(
['Salutation', 'Name', 'Sex', 'Ticket', 'Embarked', 'Cabin_Serial', 'Cabin']
Salutation
Mr        537
Miss      185
Mrs       129
Master     40
Name: count, dtype: int64
  Salutation  Salutation_Dict     Sex  Sex_Dict Embarked  Embarked_Dict Cabin_Serial
Cabin_Serial_Dict    Cabin  Cabin_Dict
0         Mr              537    male       577        S            646            C
746  B96 B98         691
1        Mrs              129  female       314        C            168            C
746      C85           1
2       Miss              185  female       314        S            646            C
746  B96 B98         691
3        Mrs              129  female       314        S            646            C
746     C123           2
4         Mr              537    male       577        S            646            C
746  B96 B98         691
   PassengerId  Survived                                                Name   Age  SibSp
...  Salutation_Dict Sex_Dict  Embarked_Dict  Cabin_Serial_Dict  Cabin_Dict
0            1         0                              Braund, Mr. Owen Harris  22.0      1
...              537      577            646                746         691
1            2         1  Cumings, Mrs. John Bradley (Florence Briggs Th...  38.0      1
...              129      314            168                746           1
2            3         1                               Heikkinen, Miss. Laina  26.0      0
...              185      314            646                746         691
3            4         1       Futrelle, Mrs. Jacques Heath (Lily May Peel)  35.0      1
...              129      314            646                746           2
4            5         0                             Allen, Mr. William Henry  35.0      0
...              537      577            646                746         691

[5 rows x 14 columns]
Cabin
A    0.466667
B    0.328338
C    0.593220
D    0.757576
E    0.750000
F    0.615385
```

```
G    0.500000
T    0.000000
Name: Survived, dtype: float64
0.3838383838383838
Index(['T', 'B', 'A', 'G', 'C', 'F', 'E', 'D'], dtype='str', name='Cabin')
{'T': 0, 'B': 1, 'A': 2, 'G': 3, 'C': 4, 'F': 5, 'E': 6, 'D': 7}
  Cabin  Cabin_Encoded
0     B              1
1     C              4
2     B              1
3     C              4
4     B              1
5     B              1
6     E              6
7     B              1
8     B              1
9     B              1
     Cabin  Cabin_Mean_Encoded
0  B96 B98            0.303907
1      C85            1.000000
2  B96 B98            0.303907
3     C123            0.500000
4  B96 B98            0.303907
5  B96 B98            0.303907
6      E46            0.000000
7  B96 B98            0.303907
8  B96 B98            0.303907
9  B96 B98            0.303907
       Survived      Died
Cabin
A      0.466667  0.533333
B      0.328338  0.671662
C      0.593220  0.406780
D      0.757576  0.242424
E      0.750000  0.250000
F      0.615385  0.384615
G      0.500000  0.500000
T      0.000000  1.000000
       Survived      Died  Prob_Ratio
Cabin
A      0.466667  0.533333    0.875000
B      0.328338  0.671662    0.488844
C      0.593220  0.406780    1.458333
D      0.757576  0.242424    3.125000
E      0.750000  0.250000    3.000000
F      0.615385  0.384615    1.600000
G      0.500000  0.500000    1.000000
T      0.000000  1.000000    0.000000
{'A': 0.875, 'B': 0.488843813387424, 'C': 1.4583333333333333, 'D': 3.125, 'E': 3.0, 'F':
1.6000000000000003, 'G': 1.0, 'T': 0.0}
  Cabin  Encode_Prob_Ratio
0     B           0.488844
1     C           1.458333
2     B           0.488844
3     C           1.458333
```

```
4     B           0.488844
5     B           0.488844
6     E           3.000000
7     B           0.488844
8     B           0.488844
9     B           0.488844
   PassengerId  Survived Salutation                                                Name
Sex  ...     Fare  Embarked  Cabin_Serial Ord_Pclass  Encode_Prob_Ratio
0            1         0         Mr                             Braund, Mr. Owen Harris
male  ...   7.2500         S             C          3           0.488844
1            2         1        Mrs  Cumings, Mrs. John Bradley (Florence Briggs Th...
female  ...  71.2833         C             C          1           1.458333
2            3         1       Miss                             Heikkinen, Miss. Laina
female  ...   7.9250         S             C          3           0.488844
3            4         1        Mrs       Futrelle, Mrs. Jacques Heath (Lily May Peel)
female  ...  53.1000         S             C          1           1.458333
4            5         0         Mr                           Allen, Mr. William Henry
male  ...   8.0500         S             C          3           0.488844
5            6         0         Mr                                 Moran, Mr. James
male  ...   8.4583         Q             C          3           0.488844
6            7         0         Mr                            McCarthy, Mr. Timothy J
male  ...  51.8625         S             E          1           3.000000
7            8         0     Master                       Palsson, Master. Gosta Leonard
male  ...  21.0750         S             C          3           0.488844
8            9         1        Mrs  Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)
female  ...  11.1333         S             C          3           0.488844
9           10         1        Mrs                Nasser, Mrs. Nicholas (Adele Achem)
female  ...  30.0708         C             C          2           0.488844

[10 rows x 14 columns]
PS C:\Users\Lenovo\AppData\Local\Programs\Microsoft VS Code>
```